TRACK 3: AUTONOMOUS AI TEAMMATES

Paytm Build for India AI Hackathon | Bengaluru Edition

Nexus Teammate

Closes tickets directly on screen instead of answering with text.

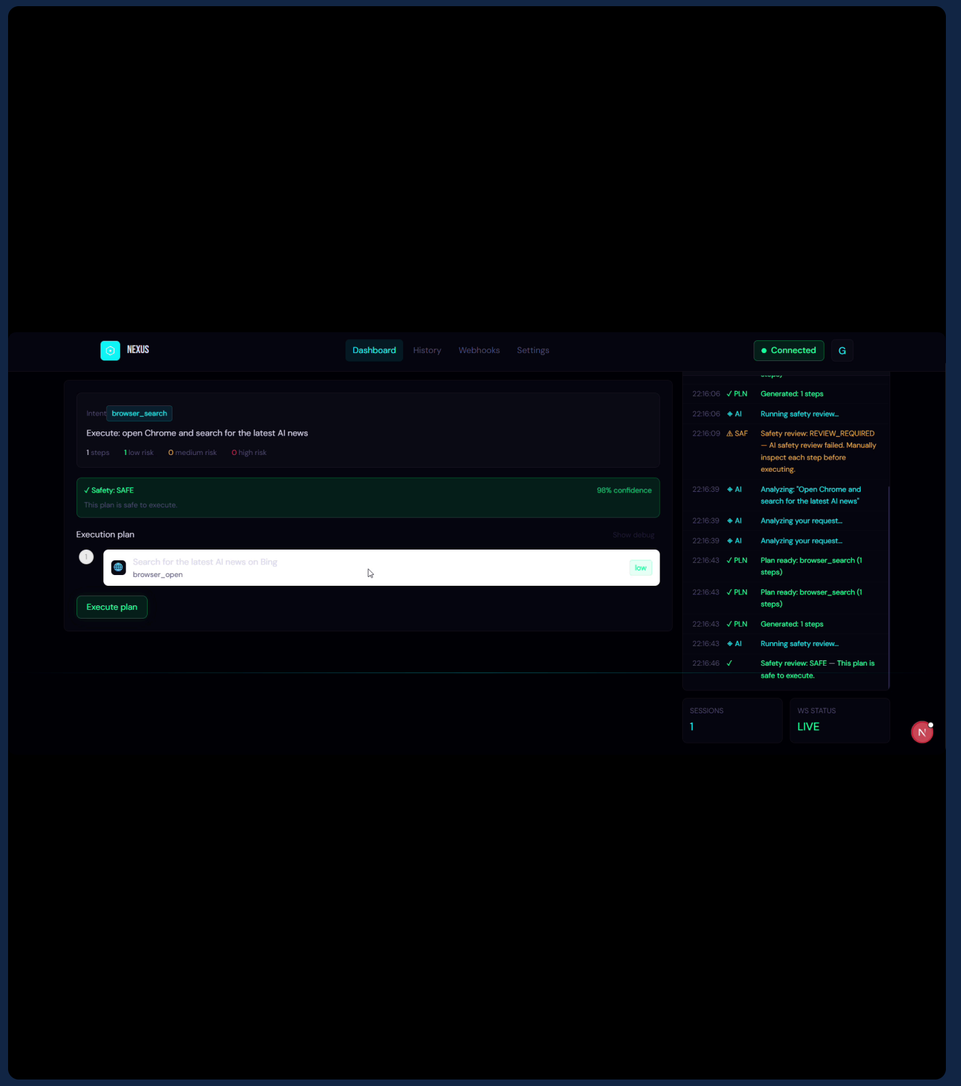
Autonomous desktop operations teammate for Paytm customer support and merchant workflows. It executes tasks across internal tools without requiring custom APIs.

106 Logged Runs
83% task success rate

7-Tier Self-Healing
Adapts when UI changes

Builder: Garv Thakre | GSoC 2026 Contributor at MetaBrainz | Built 90+ production REST APIs

Nexus Execution Engine: Live Plan & Activity Stream



Real runtime UI: generates execution plan, audits security risks, and triggers automated screen steps live.

THE REALITY IN SUPPORT OPS

Support teams spend their day clicking, not solving

THE MANUAL MAZE: ONE TRANSACTION DISPUTE TRAVERSES 5 SEPARATE SCREENS

1. Ticket Queue
Zendesk / CRM
Reads user complaint
[Human clicks here]

2. Payment Portal
Merchant Console
Looks up Txn status
[Human clicks here]

3. Bank Ledger
Banking Gateway
Verifies UTR credit
[Human clicks here]

4. Dispute Engine
Internal Web Tool
Checks refund policy
[Human clicks here]

5. Payout Desk
Finance Tool
Executes reversal
[Human clicks here]

Chatbots answer, but humans still click

Conversational AI answers queries with text, but human operators still have to manually navigate portals and update records.

Internal operations tools lack APIs

Core banking portals, legacy admin consoles, and merchant tools have no programmatic endpoints. Human screens are the only interface.

Routine disputes crowd out complex cases

Agents spend over 60% of their shifts repeating the same lookup steps instead of managing escalations and merchant relationships.

INDUSTRY REALITY: SUPPORT OVERHEAD IN DIGITAL PAYMENTS

[ADD SOURCED STAT: Indian payment gateways report up to 30% of support volume comes from transient failed payment disputes and status queries (Source: CXO Digital Pulse / Industry Average)]

AUTONOMOUS ACTION, NOT CHAT

Nexus works the ticket from open to closed

1. READ
Ingests ticket data

Extracts transaction ID, user claim, and order details.

2. PLAN
Compiles steps

Translates task into a strict multi-step browser plan.

3. REVIEW
Audits safety risks

LLM reviews plan; confirms steps before execution.

[Human in the loop: safety verdict + confirm step]

4. ACT
Operates screens

Navigates dashboards, fills forms, clicks buttons.

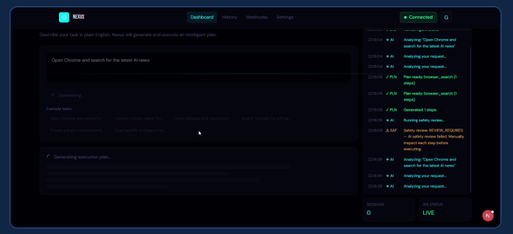
5. LOG
Records every action

Appends execution trace to immutable JSONL audit log.

TARGET WORKFLOW: FAILED-PAYMENT DISPUTE

A failed-payment dispute arrives. Nexus reads context, plans steps, reviews safety, checks status in the portal, picks refund or escalate, updates the ticket, and logs every step.

Target workflow for customer operations. Humans only handle unresolved edge cases.



CALLOUTS

1. **Strict JSON Plan:** deterministic steps.
2. **Safety Auditor:** pre-run risk verdict.
3. **Activity Log:** real-time streaming trace.

BUILT & RUNNING: Runs on a schedule (cron DB triggers) | Switches models on failure (failover routing) | Takes voice commands (browser en-IN, hi-IN)

SYSTEM ARCHITECTURE

API-first; UI automation for the gaps

TRIGGER LAYER

User Prompt (Web UI) [Built]
Plain English task entry in Electron/Next.js interface.

Scheduler Service [Built]
Cron DB stored schedules for unattended batch jobs.

Voice Input [Built]
Browser en-IN & hi-IN voice commands.

Webhooks [Built]
Trigger task runs and export status events.

INTELLIGENCE LAYER

Planner (planner.ts)
Context budgeting (12K chars); compiles high-level prompt into concrete JSON action steps.

Safety Reviewer
Independent LLM safety audit: returns SAFE, REVIEW_REQUIRED, or UNSAFE with risk flags.

Provider Failover [Built]
Automated fallback routing across Groq Llama 3.3 70B, OpenAI GPT-4o, and Anthropic Claude.

7-TIER SELF-HEALING ENGINE

Browser execution cascade (Browserengine.ts):

Tier -1: Selector Memory (learned cache)

Tier 0: Direct CSS & Site Handlers

Tier 1: DOM Fuzzy Scan (keyword scoring)

Tier 2: Accessibility Tree + Groq

Tier 3: HTML Context + LLM Re-plan

Tier 4: URL Parameter Fallback

Tier 5: Vision Fallback (Gemini Flash)

ACTIONS & AUDIT

Playwright
Browser session automation.

Pywinauto
Windows native app control.

Auto-Fix Engine
6 script repair patterns.

JSONL Trace
Step-level execution logs.

PostgreSQL
AES-256-GCM encrypted BYOK.

Core Stack: TypeScript, Node.js, Express, Playwright, Python (pywinauto), Next.js, PostgreSQL (Prisma), Electron, Docker Compose.

COMPETITIVE MOAT

It works in tools that have no API

Capability	Chatbot	API Integration	Scripted RPA	Nexus Teammate
Takes real actions across screens	No (Answers only)	Yes	Yes	**Yes**
Works without custom APIs	No	No (Requires APIs)	Yes	**Yes**
Adapts when UI layout changes	No	No	No (Brittle scripts)	**Yes (7-Tier Healing)**
Pre-execution safety review	No	No	No	**Yes (Safety verdict)**
Full step-by-step audit trace	Conversation log	Partial API logs	Local log files	**Yes (JSONL trace + DB)**
Runs unattended on a schedule	No	Yes	Yes	**Yes (Cron DB triggers)**
Switches providers on failure	No	No	No	**Yes (Automatic failover)**

THE CORE DIFFERENCE: **RPA breaks when a button moves. Chatbots stop at giving advice. Nexus adapts to screen changes and finishes the job.**

PROVEN EXECUTION & MEASURED OUTCOMES

Proven execution speed with 94.4% support accuracy

HISTORICAL RUNS (GENERAL DESKTOP & WEB TASKS)

Measured across 106 logged tasks from repository execution logs:

106
Recorded runs in log

83.0%
Fully completed tasks

5.0
Average steps per task

70.5s
Average run duration

PAYTM TRACK 3 BENCHMARK (MOCK SUPPORT CONSOLE)

18 realistic support operations (disputes, refunds, settlements, duplicate debits, KYC):

94.4%
Support task success rate (17/18)

0.90s
Average resolution time per task

7
Self-healing recovery events

0.20¢
Average cost per ticket ($0.001995)

FOR CUSTOMERS

Immediate dispute resolution with zero waiting queues.

FOR SUPPORT AGENTS

Eliminates repetitive manual portal lookups entirely.

FOR PAYTM OPERATIONS

Reduces cost per resolved ticket at scale.